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Cymbaria daurica L.: A Mongolian herbal medicine for treating eczema via natural killer cell-mediated cytotoxicity pathway

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ABSTRACT

Ethnopharmacological relevance: *Cymbaria daurica* L. (*C. daurica*) is a perennial herb known commonly as "Xinba" (Chinese) and "Kanba-Arong" (Mongolian). In Mongolia, it is used as a traditional medicine to treat eczema and other skin diseases due to its anti-swelling, anti-inflammatory, anti-hemorrhagic, and anti-itching properties. However, the potential mechanism of action for eczema treatment has not been reported.
Aim of the study: To investigate the effect of *C. daurica* on 1-chloro-2,4-dinitrobenzene (DNCB)-induced eczema in rats and the associated action mechanism.
Materials and methods: Qualitative analysis of *C. daurica* was performed by liquid chromatography-tandem mass spectrometry (LC-MS/MS). Based on information obtained from compound identification and relevant literature, the possible targets of *C. daurica* against eczema were analyzed using network pharmacology and molecular docking methods. The DNCB-induced eczema rat models were treated with different dosages of *C. daurica* extract (10, 50, and 250 mg/mL per day), and the therapeutic effects subsequently evaluated based on the degree of skin inflammation, spleen index, and hematoxylin and eosin staining (H&E staining). Enzyme-linked immunosorbent assay (ELISA), reverse transcription quantitative polymerase chain reaction (RT-qPCR), and western blotting were used to analyze the relevant target effects. The *C. daurica* mechanism of action on eczema was verified by animal experiments. High-performance liquid chromatography (HPLC) was carried out to determine the content of active ingredients in *C. daurica.* In addition, the physicochemical properties of the extract were evaluated.
Results: Our analysis of the 173 targets included in the protein–protein interaction (PPI) network identified tumor necrosis factor (TNF) and interleukin 2 (IL-2) as key targets involved in the treatment of eczema with *C. daurica* extract. Furthermore, the 173 targets were associated with the natural killer cell-mediated cytotoxicity pathway. Our results showed that *C. daurica* significantly reduced IL-2 and TNF-α serum levels in eczema rat models ($P < 0.0001$); thus, playing an important role in the anti-inflammatory response. Furthermore, according to the *p*-value, RT-qPCR and western blotting showed that the expression of Src homology 2 domain-containing protein tyrosine phosphatase 1 (SHP-1), Vav guanine nucleotide exchange factor (Vav), and growth factor receptor-bound protein 2 (Grb2) changed in the skin of the eczema model rats after treatment with the *C. daurica* extract.
Conclusion: Our study confirms that *C. daurica* can inhibit SHP-1, Vav, and Grb2 expression; thereby, inhibiting the natural killer cell-mediated cytotoxicity pathway. These results provide insight into the mechanism of *C. daurica* in treating eczema.

Abbreviations: C. daurica, Cymbaria daurica L.; DNCB, 1-chloro-2,4-dinitrobenzene; LC-MS/MS, liquid chromatography-tandem mass spectrometry; HPLC, high-performance liquid chromatography; H&E staining, hematoxylin and eosin staining; ELISA, enzyme-linked immunosorbent assay; RT-qPCR, reverse transcription quantitative polymerase chain reaction; PPI, protein–protein interaction; TNF, tumor necrosis factor; IL-2, interleukin-2; SHP-1, Src homology 2 domain-containing protein tyrosine phosphatase 1; Vav, Vav guanine nucleotide exchange factor; Grb2, growth factor receptor-bound protein 2; UAE, ultrasound-assisted extraction; MS, Mass spectrometry; PBS, phosphate-buffered saline; KEGG, Kyoto Encyclopedia of Genes and Genomes; SDS-PAGE, sodium dodecyl sulfate-polyacrylamide gel electrophoresis; PVDF, polyvinylidene difluoride; DL, drug-likeness; AE, atopic eczema.

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1. Introduction

Eczema is an inflammatory skin disease, which can be classified as acute, subacute, and chronic based on disease progression (Li and Liu, 2011). It is often characterized by skin pruritus and redness, skin infiltration, edema, papulovesicles accompanied by dryness, scales, and hyperkeratosis (Hanifin, 2008; Chang et al., 2007; Coenraads, 2012), epidermal barrier dysfunction, and various other symptoms (Sohn et al., 2011). The incidence of eczema is high, and long-term recurrence reduces patient quality of life considerably (Langan et al., 2020). In addition, eczema in specific body parts can cause anxiety; thereby, significantly impacting both the physical and mental health of patients (Wang, 2022). The annual prevalence rate of hand eczema was 9.7% according to Swedish population-based data from 1996, and a recent study in Norway showed that the lifetime prevalence rate of hand eczema was 11.3% (Agner and Elsner, 2020). Currently, there are no specific drugs or methods available for effective treatment of this disease. Antihistamine drugs, calcium agents, vitamins, antibiotics, and hormones (e.g., glucocorticoids) can temporarily alleviate clinical symptoms; however, their long-term use can lead to skin lesions, wounds prone to infection, skin atrophy, and other adverse reactions (Wollenberg et al., 2016). For example, antihistamine therapy may reduce a child's learning ability and attention, and some could also have a central inhibitory effect. Therefore, the use of antihistamines to treat eczema in children, pregnant and lactating women, and the elderly is not recommended (Li, 2021). Hormone treatment may induce side effects such as infection, skin atrophy, and abnormal skin pigmentation (Zhang, 1981). Azathioprine is an immunomodulator that inhibits the synthesis of purines and immunoglobulins, which can lead to accumulation of 6-thioguanine nucleotides in lymphocytes; thereby, inhibiting the expression of endogenous inflammatory mediators (Thomas et al., 2005). According to previous studies, azathioprine treatment was discontinued due to adverse effects such as gastrointestinal symptoms, influenza-like symptoms, and elevated liver enzymes (Oosterhaven et al., 2017). In traditional Chinese medicine, plants such as *Vigna umbellate*, *Forsythia suspensa*, and *Nepeta cataria* L. (Lu and Zhu, 1992; Xu and Gao, 2018; Yu, 2022) have been used to treat eczema. Despite the various methods of treatment available to patients, the effects are not optimal. For hand eczema, the average annual cost per patient is between €1712 and €9, 792, with a high recurrence rate of 65% (Veien et al., 2008). Furthermore, previous eczema patient feedback has revealed that only 15% of patients were highly satisfied with the treatment effect, and 26% found it unsatisfactory (Lee et al., 2019); thus, highlighting the need for safer and more effective drugs for treatment of eczema.

As an important part of the world's traditional culture, Mongolian medicine is a traditional medicine with regional and ethnic characteristics with an important role in the world's medical system (Song et al., 2022). Mongolian medicine is the wisdom crystallization formed during the struggle of Mongolian people against diseases (Bai and Fu, 2022). Mongolian medicine not only has a deep historical origin with traditional Chinese medicine, but has also absorbed the essence of ancient Tibetan and Indian Ayurvedic medicine during its development (Long et al., 2022). According to literature reports, there are more than 2000 varieties of Mongolian medicine in China, which may play an important role in public health fields such as anti-tumor, immunomodulatory, bacteriostatic and cardiovascular and cerebrovascular protection (Wang et al., 2017; Li et al., 2019; Zhou et al., 2022).

Cymbaria daurica L. is a perennial herb of the Orobanchaceae family (Fig. 1). As a traditional Mongolian medicine, the local name of *C. daurica* is "Xinba" (Chinese) and "Kanba-Arong" (Mongolian). The plant can reach ~23 cm in height, displaying yellow flowers and white linear or linear lanceolate leaves (http://www.iplant.cn/info/Cymbaria%20daurica). It commonly occurs on steep mountain slopes growing close to the ground. The earliest record of *C. daurica* appears to come from a work *Ren Yao Bai Jing Jian* (认药白晶鉴) written in Qing Dynasty (A.D. 1636–1912). Subsequently, the folk and traditional medicinal value of *C. daurica* has been detailed in many medical works, as shown in Table 1 and Fig. 2. Medicinal uses include heat dissipating and blood coagulating effects, in addition to pain, itching and tumescence relief. It is mainly used to treat various skin diseases by decocting soup or frying carbon for external use. At present, it is still widely used to treat eczema and dermatitis in Xilingol League and other places in Inner Mongolia (Ge et al., 2019). In addition, *C. daurica* can also be used in compatibility in Mongolian medicine. For example, it can be used together with other herbs to treat fever and relieve pain by forming the "Ba Wei Xin Ba San". In modern pharmacological research, *C. daurica* can be used as a diuretic and has anti-inflammatory, anti-oxidant, anti-tumor, and anti-bacterial properties (Gong et al., 2020b; Shi et al., 2020). Based on a study by Guo et al. the 70% ethanol extract of *C. daurica* has anti-inflammatory and analgesic effects, which can reduce ear swelling in mice, foot swelling in rats, and the pain threshold of mice (Guo et al., 2017). However, the specific biological mechanisms involved in treating eczema remain unclear.

In this study, we aimed to elucidate the immunomodulatory

Table 1
The medical works describing *Cymbaria daurica* L.

No	Source	Written Time
1	*Ren Yao Bai Jing Jian (认药白晶鉴)*	In the middle of the 18th century
2	*Ren Yao Xue (认药学)*	In the late of the 18th century
3	*Meng Yao Zheng Dian (蒙药正典)*	In the early 19th century
4	*Nei Meng Gu Zhong Cao Yao (内蒙古中草药)*	In 1972 A.D.
5	*Nei Meng Gu Meng Yao Cai Biao Zhun (内蒙古蒙药材标准)*	In 1987 A.D.
6	*Nei Meng Gu Zhi Wu Zhi (内蒙古植物志)*	In 1989 A.D.
7	*Zhong Huang Ben Cao-Meng Yao Juan (中华本草-蒙药卷)*	In 2004 A.D.
8	*Yin Shan Zhong Meng Yao Zi Yuan Tu Zhi (阴山中蒙药资源图志)*	In 2020 A.D.

Fig. 1. Morphology of *Cymbaria daurica* L. (A) Fresh whole plants; (B) Whole plant specimen.

mechanism of *C. daurica* in the treatment of eczema. First, we identified the plant's chemical components via liquid chromatography-tandem mass spectrometry (LC-MS/MS). Second, we analyzed the potential targets through network pharmacology and additional methods. Last, we verified our results with animal experiments. Our results provide a research basis for the targeted treatment of eczema, offering modern pharmacological validation of the efficacy of traditional Mongolian medicine.

2. Materials and methods

2.1. *C. daurica* and sample preparation

C. daurica samples were collected in May 2019, from grasslands (44°34′5.8″ N, 117°35′6.7″ E, Elevation: 1058.9 m) located in Haoletugaole town in the western region of Ujimqin Banner, Xilin Gol League, Inner Mongolia Autonomous Region, China. The annual mean temperature was approximately ~1.70 °C (https://www.worldclim.org/). It was certified by Professor Chunhong Zhang of Baotou Medical College. The collection, identification, and preservation methods of *C. daurica* are in accordance with the Technical Specifications for the National Survey of Traditional Chinese Medicine Resources (Huang and Wang, 2015). A voucher specimen was deposited in the Herbarium of HIMC (Specimen No.: 152500180629025LY) for future reference. The 70% ethanol extract of *C. daurica* was prepared as follows, based on the procedure used by Gong (2020a) and Guo et al. (2017). The whole grass of *C. daurica* (30 g) was dried, crushed, and added to 70% ethanol (solid: liquid = 1:10) in a round bottom flask. The flask was placed in an electric heating sleeve and heated to a boil. The temperature was adjusted to maintain a uniform flow rate for 2 h. Then, the solution was filtered using a Büchner funnel, and the residue was reheated and allowed to reflow for 2 h, according to the aforementioned method. Extraction was performed twice, and the filtered liquid was combined. After concentration under reduced pressure, the concentrated extract was dried. The 70% ethanol extract of *C. daurica* was stored at 4 °C for future use. Solutions containing 10, 50, and 250 mg/mL were prepared for animal experiments (Hong et al., 2020).

2.2. Qualitative determination of *C. daurica* chemical compositions

The 70% ethanol extract of *C. daurica* was weighed, of which 50 mg was added to a 5 mL volumetric flask and dissolved in methanol for ultrasound-assisted extraction (UAE) (40 kHz, 30 min). The solution was diluted with methanol to a fixed volume. Subsequently, the solution was taken, centrifuged (6000 rpm, 4 °C, 5 min), and filtered through a microporous membrane (0.22 μm) in preparation for LC-MS/MS

Fig. 2. Some ancient medical literatures describing *C. daurica*.

analysis.

Chromatographic analyses of the extracted compounds were performed using a SCIEX X500 QTOF system (SCIEX, Framingham, MA, USA) with an autosampler. An ACQUITY UPLC BEH C18 chromatography column (100 mm × 2.1 mm, 1.7 μm) from Phenomenex (Torrance, CA, USA) was used for chromatographic separation. The mobile phase comprised of water containing 0.1% formic acid (solvent A) and acetonitrile (solvent B). Gradient elution was set as follows: 0–72 min, 5%–95% B. The flow rate was set to 0.3 mL/min, and the injection volume was 5 μL. The column temperature was 40 °C.

Mass spectrometry (MS) data were captured using the SCIEX X500 QTOF equipped with an electrospray ionization interface in the negative ion modes. The following parameters were selected: temperature, 500 °C; spray voltage, −4500V (negative ion modes); curtain gas, 30 psi; and CAD gas, 7 psi. Full-scan MS data were generated, with masses ranging between 100 and 1500 Da. The step normalized collision energy setting was −20, −40, and −60 V (negative ion modes). All MS spectra were analyzed using the SCIEX OS software.

2.3. Reagents

DNCB was purchased from MREDA (USA). IL-2 and TNF-α ELISA kits were purchased from Shanghai Hengyuan Biological Technology Co., Ltd. (China). Phosphate-buffered saline (PBS), Tris-buffered saline with Tween (10 ×), Tris-glycine running buffer (5 ×), Western blotting transfer buffer (10 ×), non-fat powdered milk, SDS-PAGE Gel Kits, BCA kits, and high-efficiency RIPA buffer were purchased from Solarbio (China). PrimeScript™ RT reagent kits and TB Green ® Premix Ex Taq ™ II were purchased from TaKaRa Biotechnology Co. (Japan). Antibodies against Src homology 2 domain-containing protein tyrosine phosphatase 1 (SHP-1), Vav guanine nucleotide exchange factor 1 (Vav1), growth factor receptor-bound protein 2 (Grb2), and GAPDH were purchased from Cell Signaling Technology (USA). Horseradish peroxidase (HRP)-conjugated affinity-pure goat anti-rabbit IgG (H + L) was purchased from Beyotime (China).

2.4. Network pharmacology and molecular docking

2.4.1. Collection of bioactive compounds and related targets

We collated the chemical constituents from our LC-MS/MS results and related literature (Dai et al., 2002; Bai et al., 2018). Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform (TCMSP; http://tcmspw.com/) (Zhu et al., 2021) was used to screen the chemical components of *C. daurica.* Since the animal experiments in this study involved transdermal drug delivery, the ingredients meeting the criteria of drug-likeness (DL) ≥ 0.18 were selected as the potential active ingredients. The selected compounds were found in PubChem (https://pubchem.ncbi.nlm.nih.gov/) to obtain the SMILES number. The corresponding targets of these compounds were searched in SwissTargetPrediction (http://www.swisstargetprediction.ch/).

2.4.2. Determination of eczema disease and inflammation targets

Using "eczema" and "inflammation" as keywords, we searched the GeneCards (http://www.genecards.org/), OMIM database (http://www.omim.org) and DisGeNET databases (http://www.disgenet.org/) to obtain eczema and inflammation-related targets.

2.4.3. Construction of the compound compositions–target network

The target genes of *C. daurica* against inflammation and eczema were obtained by the intersection using the jvenn online database (http://jvenn.toulouse.inra.fr/app/index.html). The candidate targets of *C. daurica* used in inflammation and eczema therapy were predicted. The STRING database (https://www.string-db.org/) was used to construct a protein–protein interaction (PPI) regulation network, which was visualized using Cytoscape 3.7.1 software. Subsequently, a network of *C. daurica* ingredient compounds-inflammation and eczema targets was constructed.

2.4.4. Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis

To elucidate the possible pathways involved in the treatment of eczema with *C. daurica,* Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis ($P < 0.05$) was carried out using the DAVID database (https://david.ncifcrf.gov/).

2.4.5. Molecular docking

We searched the Uniprot database (https://www.uniprot.org/) for three receptors in the natural killer cell-mediated cytotoxicity pathways and downloaded their 3D structures from the RCSB PDB database (https://www.rcsb.org/). The AutoDock 1.5.6 software (Morris et al., 2009) was used to remove water molecules and save the PDBQT files. The molecular ligand 2D structures were downloaded using the PubChem database. Chem3D software was used to transform the 2D structures into 3D structures, which were saved as a docking ligand in the PDBQT format in AutoDock Tools 1.5.6. AutoDock Vina 1.5.6 was used to dock a small molecule ligand with a large molecule receptor. When the binding energy was ≤ -5.0 kcal mol^{-1}, the compound was considered to have a high affinity for the targets. The conformation with the lowest score was selected for visualization in PyMOL 2.0 (https://pymol.org/2/).

2.5. Animal experiments

2.5.1. Experimental animals

Male Sprague-Dawley rats (n = 42, 200–250 g) were purchased from SPF Biotechnology Co., Ltd. (Beijing, China; License No. SCXK[Jing] 2010-0016). Experiments were conducted in accordance with the ethical guidelines for the care of laboratory animals and approved by the animal experimental ethics review committee of Baotou Medical College (approval number: Baotou Medical Lun Audit 2022 No. 095). The rats were kept at the animal house in standard laboratory conditions (temperature, 25 ± 2 °C; relative humidity, 55 ± 5%; 12 h light-dark cycle) for one week prior to the experiment for acclimation. The rats were provided with commercial feed and water ad libitum unless otherwise stated. Cage bedding was replaced daily. The experiment was conducted after the health status of the animals was confirmed.

2.5.2. Establishment of the eczema model and administration

The rats were separated into six groups (n = 7, each group): blank control group, model group, positive drug group, along with low-, medium-, and high-dose *C. daurica* groups. Hair was shaved from the left dorsal area (area: 2 × 2 cm^2) and right dorsal area (4 × 4 cm^2) using an electric razor. Rats were sensitized for the first five days of the experiment using 100 μL of 7% DNCB solution (dissolved in acetone:olive oil, 3:1 solution) administered via the left dorsal shaved skin area. From days 6–8, 200 μL of 2.5% DNCB solution or vehicle was applied via the right dorsal shaved skin area for 6 h. After exposure, rat skin condition was observed and photographed (Canon 80D, Canon, Japan).

Skin condition was evaluated using the following criteria: Erythema and eschar: 0, (no response); 1, slight erythema (barely visible); 2, obvious erythema (scattered or small patches of erythema); 3, moderate to severe erythema; 4, severe erythema (purplish red) to slight eschar formation. Edema: 0, no edema; 1, slight edema (barely visible); 2, moderate edema (clear outline of skin bulge); 3, severe swelling (skin bulge of ∼≥1 cm). Animals that showed a skin reaction >2 were deemed to have a skin sensitization reaction, based on the test method of skin sensitization for chemicals (National Standard of the People's Republic of China).

Treatment began one day after successful modeling had been demonstrated. Varying concentrations of the 70% ethanol extract of *C. daurica* (10, 50, and 250 mg/mL) and positive drug (mometasone furoate cream, 0.2 g each time; Li and Li, 2013) were administered daily for 14 days. The concentration of the mometasone furoate cream was 0.1% (10 g of mometasone furoate cream contained 10 mg of the effective drug, approval number 'H20050610'). The control and model groups were not treated. Blood was collected after 14 days from the abdominal aorta, from which blood serum was subsequently extracted. Skin samples were collected; some were used for histological analysis, and the rest were snap-frozen in liquid nitrogen and stored at −80 °C until required.

2.5.3. Determination of the spleen index

The spleen is an important immune organ of the body, and the enlargement of the spleen indicates activation of the immune response in diseases (Jabeen et al., 2020). In the study of skin diseases, the use of DNCB can over-regulate the immune response and increase the spleen index. Therefore, spleen index was selected as a reference index in this study (Fan et al., 2021). After the experimental period, the rats were weighed, and their spleens were removed and weighed. The spleen index was calculated as follows: spleen index = [rat spleen weight (mg)]/[experimental rat weight (g)] (Lv et al., 2022). The spleen index of the treatment groups was compared with that of the control group, and the differences were calculated.

2.6. Hematoxylin and eosin staining (H&E)

H&E staining is an important method of immunohistochemical analysis, which can be used to evaluate the structural changes of skin tissue (Song et al., 2021). Therefore, we selected H&E staining for rat skin tissue analysis. Skin tissues were fixed in 4% paraformaldehyde and paraffin-embedded. Subsequently, 4-μm-thick sections were cut with a microtome. H&E staining was performed according to the manufacturer's instructions. Histopathological examination was performed under a light microscope (Leica DM2000, Leica, Germany).

2.7. Enzyme-linked immunosorbent assay

The circulating concentration of proteins or polypeptides in the body plays a key role in maintaining the metabolic homeostasis of the body. As this concentration is affected by various pathophysiological events, it can be used to help diagnose or monitor various diseases (Aydin, 2015). Antigen-antibody reaction is sensitive and has a wide range of applications (Klauenberg et al., 2015). All techniques that use enzymes to display antigen-antibody reactions are commonly referred to as the enzyme-linked immunosorbent assay method (Aydin, 2015). Therefore, this study used the ELISAs method to detect indexes in rat serum. Tumor necrosis factor α (TNF-α) and interleukin 2 (IL-2) kits were used to measure relative levels in the rat serum, following the manufacturer's instructions.

2.8. Reverse transcription quantitative polymerase chain reaction

RT-qPCR is an important tool for quantifying gene expression levels and the threshold cycle (C_T) is a key quantitative indicator. The comparative C_T method, also known as $2^{-\triangle\triangle CT}$ method, is commonly used for presenting quantitative real-time PCR data. It is user friendly and can present the data as a "fold change" in expression, which is suitable for comparing gene expression levels in different samples (Schmittgen et al., 2008). The $\triangle C_T$ values were calculated using the following equation:

$\triangle C_T = C_{T\ (sample)} - C_{T\ (endogenous\ control)}$, $\triangle\triangle C_T = \triangle C_{T\ (sample)} - \triangle C_{T\ (untreated)}$, and fold change $= 2^{-\triangle\triangle CT}$

Total RNA was extracted from rat skin tissues using TRIzol reagent and reverse-transcribed using the PrimeScript™ RT Master kit, according to the manufacturer's instructions. Subsequently, cDNA was subjected to reverse transcription quantitative polymerase chain reaction (RT-qPCR) under the following conditions: 40 cycles at 95 °C for 30 s, primer annealing at 95 °C for 5 s, and extension at 60 °C for 34 s. Specific primer sequences are listed in Table 2. All primers were evaluated, the fluorescence signal was recorded (Applied Biosystems, Waltham, MA, USA), and the relative values were compared to those of the control group.

2.9. Western blotting

Western blotting is one of the commonly used techniques in molecular biology and proteomics (Mishra et al., 2017), which can be used to detect the expression level of target protein in samples (Taylor and Posch, 2014). In this study, Western blotting was used to verify the predicted targets and related mechanisms. The proteins in the skin tissues of rats from each group were extracted using high-efficiency RIPA buffer, and the concentrations of proteins were quantified by BCA kits. Equal amounts of protein samples (40 μg) were separated on 12% sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and subsequently transferred to polyvinylidene difluoride (PVDF) membranes. The membranes were blocked in 5% non-fat milk for 1 h before incubating with the respective primary antibodies: anti- Grb2. (Cat. No.: 36344S, 1:1000), anti-Vav1 (Cat. No.: 2502S, 1:1000), anti-SHP-1 (Cat. No.: 26516S, 1:1000), and anti-GAPDH (Cat. No.: 2118S, 1:2000) at 4°C overnight. The membranes were then washed thrice for 10 min each with TBST. Subsequently, the membranes were incubated using corresponding secondary antibodies: horseradish peroxidase (HRP)-conjugated affinity-pure goat anti-rabbit IgG (H + L, 1:2000). Signal strength was measured by Western Bright™ ECL spray and quantitatively assessed by Syngene GeneTools software (UK).

2.10. Quantitative determination of the chemical composition of *C. daurica*

The 70% ethanol extract of *C. daurica* was analyzed using high-performance liquid chromatography (HPLC). The Select Waters HSS T3 column (250 mm × 4.6 mm, 5 μm, Milford, MA, USA) was used for the analysis of verbascoside by HPLC. The mobile phase consisted of water (solvent A) and acetonitrile (solvent B). The gradient was set as follows: 0–25 min, 5–70% B. The flow rate was set to 1 mL/min and the injection volume was 5 μL. The temperature of the column was maintained at 30°C and spectrophotometric detection was carried out at a wavelength of 330 nm.

2.11. Moisture and ash content determination of the *C. daurica* extract

Moisture determination of the extract was performed in accordance with the Pharmacopoeia of the People's Republic of China 0832 moisture determination method, repeated thrice. The 70% ethanol extract of *C. daurica* was crushed and filtered through a No. 2 sieve. The extract was weighed (2 g) and spread at a thickness of 5 mm in a flat weighing bottle. The bottle cap was opened and dried for 5 h at 100 °C, and placed back on the bottle. The bottle was placed in the dryer, cooled for 30 min, and accurately weighed. It was dried for a second time at 100 °C for 1 h, cooled, and weighed until the difference between two consecutive weights was no more than 5 mg. The water content (%) of the 70% ethanol extract of *C. daurica* was calculated according to the weight loss.

For the ash content determination, 2 g of the 70% ethanol extract of *C. daurica* was placed in the crucible and slowly heated (paying attention to avoid burning) until complete carbonization. Then, the temperature was gradually increased to 600°C, until the sample completely turned to ash and constant weight was obtained by precision weighing. The ash product was dried at 600 °C for 1 h, cooled, and weighed until the difference between two consecutive weights was no more than 5 mg. The total ash content (%) in the 70% ethanol extract of *C. daurica* was calculated according to the residue weight.

Table 2
Primer sequences of target genes and internal reference (GAPDH).

Gene	Forward	Reverse
Grb2	CGTGTCCAGGAACCAGCAGATATTC	AATGAAGTCTCCTCGGCGAAAGC
Vav3	ATTGCCATCGCTCGGTATGACTTC	GCCCACCCTGCCATTTACTTCTC
SHP-1	CGGCACCATCATCCACCTCAAG	ACTCTCACGCACAAGAAATGTCCAG
GAPDH	GCACAGTCAAGGCTGAGAATG	ATGGTGGTGAAGACGCCAGTA

2.12. Statistical analysis

All data were analyzed using GraphPad Prism 9 (GraphPad Software, San Diego, CA, USA). The results were presented as the mean ± standard deviations of three independent experiments, and $P < 0.05$ was considered statistically significant.

3. Results

3.1. Chemical compositions analysis of *C. daurica*

A total of 39 chemical compounds were identified in the 70% ethanol extract of *C. daurica* from LC-MS/MS (Table 3). Eighteen of these compounds have been reported in the literature. We preliminarily determined the chemical compositions from data such as retention time, major fragments, and ionic species. Based on these analyses, the compounds were categorized into four main categories: phenylethanoid glycosides, flavonoids, iridoids, and phenolic acids. A number of these compounds were identified using standard substances. The total ion chromatogram of *C. daurica* in negative ion mode is shown in Fig. 3. The data on the 39 identified compounds are listed in Table 3.

3.2. Network pharmacology

3.2.1. Active components of *C. daurica*

Ninety-eight compounds were selected using LC-MS and literature analysis. These were imported into the TCMSP database and 31 chemical components were screened based on the criterion that DL $\geq$ 0.18 (Table 4).

3.2.2. Overlapped targets analysis

By deleting duplicate values and converting standard gene symbols, 700 targets associated with *C. daurica* were identified from TCMSP and Swiss Target Prediction databases. A total of 3310 targets associated with eczema and 5720 targets associated with inflammation were obtained from the GeneCards, OMIM and DisGeNET databases. After matching the targets of compounds with the targets of eczema and inflammation, the intersection was determined, leading to 173 common

Table 3
C. daurica chemical compounds identification.

NO.	RT (min)	Adducts	Formula	Mass (m/z)	Molecular weight	Name
1	0.85	[M-H]⁻	$C_{17}H_{24}O_{10}$	387.130	388.4	Geniposide
2	1.44	[M-H]⁻	$C_{15}H_{22}O_{10}$	361.114	362.33	Catalpol
3	2.23	[M-H]⁻	$C_{15}H_{22}O_{9}$	345.119	346.33	Aucubin
4	2.76	[M-H]⁻	$C_{16}H_{22}O_{10}$	373.114	374.34	Geniposidic acid
5	3.61	[M-H]⁻	$C_{15}H_{24}O_{9}$	347.135	348.34	Ajugol[a]
6	4.69	[M-H]⁻	$C_{16}H_{24}O_{10}$	375.130	376.36	Loganic acid
7	5.28	[M-H]⁻	$C_{21}H_{28}O_{13}$	487.146	488.4	Cistanoside F
8	6.32	[M-H]⁻	$C_{25}H_{28}O_{16}$	583.130	584.5	Neomangiferin
9	7.71	[M-H]⁻	$C_{16}H_{20}O_{9}$	355.103	356.32	Gentiopicroside[a]
10	11.18	[M-H]⁻	$C_{29}H_{36}O_{15}$	623.198	624.6	Verbascoside
11	11.91	[M-H]⁻	$C_{29}H_{36}O_{15}$	623.198	624.6	Isoacteoside
12	11.95	[M-H]⁻	$C_{35}H_{46}O_{20}$	785.251	786.7	Echinacoside[a]
13	12.18	[M-H]⁻	$C_{27}H_{30}O_{16}$	609.146	610.5	Rutin
14	12.29	[M-H]⁻	$C_{34}H_{44}O_{19}$	755.240	756.7	Forsythoside B
15	13.56	[M-H]⁻	$C_{27}H_{34}O_{14}$	581.188	582.5	Naringin dihydrochalcone
16	13.85	[M-H]⁻	$C_{27}H_{30}O_{15}$	593.151	594.5	Lonicerin
17	14.82	[M-H]⁻	$C_{21}H_{20}O_{10}$	431.098	432.4	Genistin
18	15.50	[M-H]⁻	$C_{30}H_{38}O_{15}$	637.214	638.6	Leucosceptoside A
19	15.85	[M-H]⁻	$C_{21}H_{20}O_{11}$	447.093	448.4	Cynaroside
20	16.44	[M-H]⁻	$C_{31}H_{38}O_{17}$	681.204	682.6	Tenuifoliside A
21	17.86	[M-H]⁻	$C_{18}H_{19}NO_{4}$	312.124	313.3	Norisoboldine
22	18.4	[M-H]⁻	$C_{31}H_{40}O_{15}$	651.229	652.6	Martynoside
23	18.97	[M-H]⁻	$C_{15}H_{10}O_{7}$	301.035	302.23	Quercetin
24	19.03	[M-H]⁻	$C_{15}H_{10}O_{6}$	285.040	286.24	Luteolin
25	19.48	[M-H]⁻	$C_{16}H_{12}O_{7}$	315.051	316.26	Isorhamnetin
26	19.60	[M-H]⁻	$C_{31}H_{40}O_{15}$	651.229	652.6	Cistanoside D
27	22.01	[M-H]⁻	$C_{15}H_{10}O_{5}$	269.046	270.24	Apigenin
28	22.94	[M-H]⁻	$C_{16}H_{12}O_{6}$	299.056	300.26	Diosmetin
29	22.97	[M-H]⁻	$C_{17}H_{14}O_{7}$	329.067	330.29	Tricin
30	29.13	[M-H]⁻	$C_{15}H_{10}O_{4}$	253.050	254.24	Chrysophanol
31	35.14	[M-H]⁻	$C_{30}H_{48}O_{5}$	487.343	488.7	Asiatic acid
32	40.38	[M-H]⁻	$C_{33}H_{58}O_{14}$	677.375	678.8	Gingerglycolipid B
33	48.32	[M-H]⁻	$C_{20}H_{28}O_{2}$	299.202	300.4	Vitamin A acid
34	52.76	[M-H]⁻	$C_{14}H_{28}O_{2}$	227.202	228.37	Myristic acid
35	54.28	[M-H]⁻	$C_{30}H_{48}O_{3}$	455.353	456.7	Betulinic acid
36	54.47	[M-H]⁻	$C_{36}H_{58}O_{9}$	633.401	634.8	Ecliptasaponin
37	56.65	[M-H]⁻	$C_{18}H_{32}O_{2}$	279.233	280.4	Linoleic acid
38	58.04	[M-H]⁻	$C_{30}H_{46}O_{3}$	453.330	454.7	Betulonic acid
39	62.51	[M-H]⁻	$C_{30}H_{48}O_{4}$	471.348	472.7	Hederagenin

[a] Compounds were identified using standard substances, the remaining compounds were identified by fragmentation regularities.

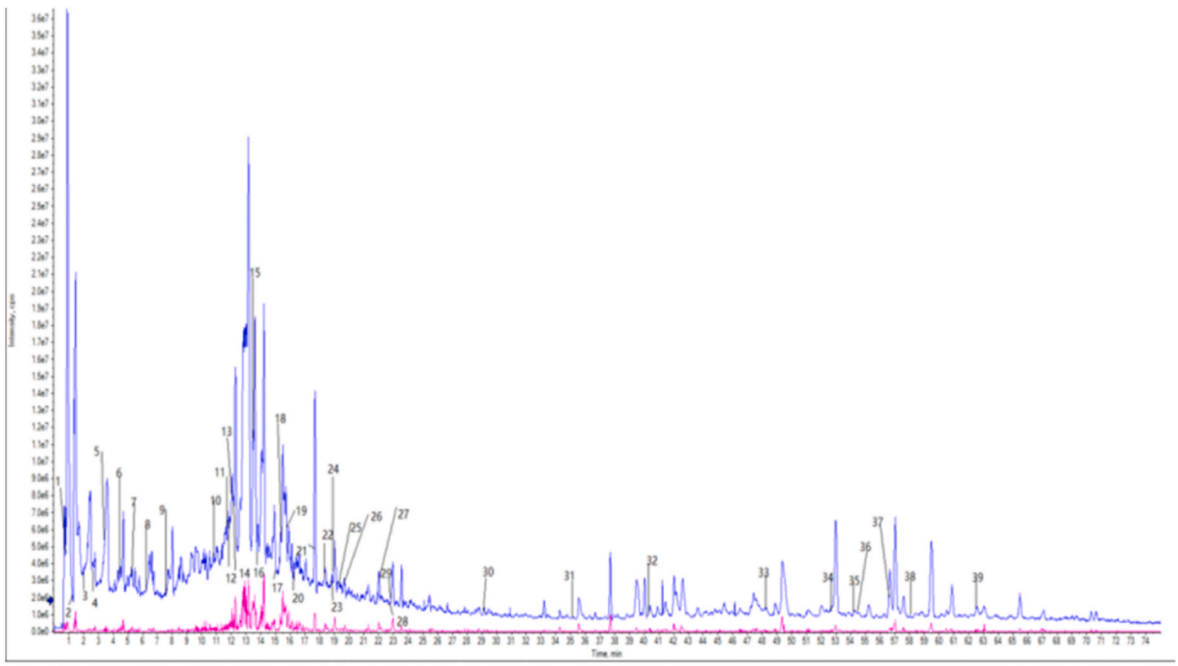

Fig. 3. The total ion chromatogram (TIC) results of the 70% ethanol extract of *C. daurica* in negative mode.

target genes being identified, as shown in Fig. 4A.

3.2.3. Component target-disease targets network analysis

The component targets–disease targets network was constructed using 173 intersection targets and 31 active components by Cytoscape 3.7.1., including 206 nodes and 931 edges (Fig. 4B). The top three compounds with high degree values were echinacoside, linolenic acid ethyl ester, and diisooctyl phthalate.

Table 4
Chemical constituents of *C. daurica*.

NO.	Name	MW	DL
1	2-Methoxy-4-vinylphenol	150.17	0.85
2	Lup-20(29)-en-3-one	424.7	0.78
3	Cynaroside	448.4	0.78
4	Cistanoside F	488.4	0.69
5	Rutin	610.5	0.68
6	Jionoside D	638.6	0.68
7	Isoacteoside	624.6	0.66
8	Verbascoside	624.6	0.62
9	Cistanoside D	652.6	0.59
10	Martynoside	652.6	0.58
11	Vitamin E	430.7	0.55
12	Catalpol	362.33	0.44
13	Geniposide	388.4	0.44
14	(E,E,E,E)-squalene	410.7	0.42
15	Geniposidic acid	374.34	0.41
16	Loganic acid	376.36	0.40
17	Gentiopicroside	356.32	0.39
18	Diisooctyl phthalate	390.6	0.39
19	Echinacoside	786.7	0.38
20	Octacosane	394.8	0.37
21	Heptacosane	380.7	0.36
22	Tricin	330.29	0.34
23	Aucubin	346.33	0.33
24	Ajugol	348.34	0.32
25	Quercetin	302.23	0.28
26	Tritetracontane	605.2	0.27
27	Chrysoeriol	300.26	0.27
28	Luteolin	286.24	0.25
29	Kaempferol	286.24	0.24
30	Apigenin	270.24	0.21
31	Linolenic acid ethyl ester	306.5	0.20

3.2.4. PPI network analysis

The 173 related targets were used as input in STRING 11.5 to construct a PPI network. There were 173 nodes and 1841 edges in the network diagram. The results input into Cytoscape 3.7.1 for analysis were shown in Fig. 4C. The line thickness represents the confidence in the reported association. The number of connections of a node represents the degree of the node. The higher the degree, the greater the position of the gene in the regulatory network, indicating that it is a central gene. According to the topological network analysis results, the top ten core genes were *TNF, AKT1, ALB, TP53, VEGFA, EGFR, SRC, STAT3, PTPRC,* and *IL-2*, thereby indicating that they may play a key role in treating eczema.

3.2.5. KEGG enrichment analysis

The KEGG pathway analysis showed 34 enriched pathways ($P < 0.05$) in the DAVID database, mainly associated with cancer, immunity, inflammation, cell regulation, and metabolism. The top 15 KEGG pathways were related to eczema with significant *P*-values (Fig. 4D). Pathways in cancer, hepatitis B, Th17 cell differentiation, bladder cancer, and natural killer cell-mediated cytotoxicity were the top five enriched pathways. Since eczema is an inflammatory skin disease, the corresponding natural killer cell-mediated cytotoxic pathways were selected for additional analysis. Based on the signaling pathway map, we selected closely related targets in key positions: SHP-1, Vav, and Grb2 to verify the therapeutic effects of the C. daurica extract on eczema in rats.

3.2.6. Molecular docking

The molecular docking was used to verify the binding activity of three targets in the natural killer cell-mediated cytotoxicity pathway (SHP-1, Vav3, and Grb2) and five compounds (echinacoside, linolenic acid ethyl ester, diisooctyl phthalate, and ajugol, gentiopicroside) in *C. daurica* identified by LC-MS/MS and network pharmacology analyses. The higher the binding activity between the compound and the protein receptor, the lower the binding energy. In this study, the docking results of the five core chemical components and three target receptors showed that the binding energies were almost ≤ -5.0 kcal mol^{-1}. We can conclude that all three targets had strong binding affinity to echinacoside (Fig. 5A). Moreover, the targets and the compound are linked to a variety of amino acids (such as ASP-133, ASN129, GLU-80, GLY143, and

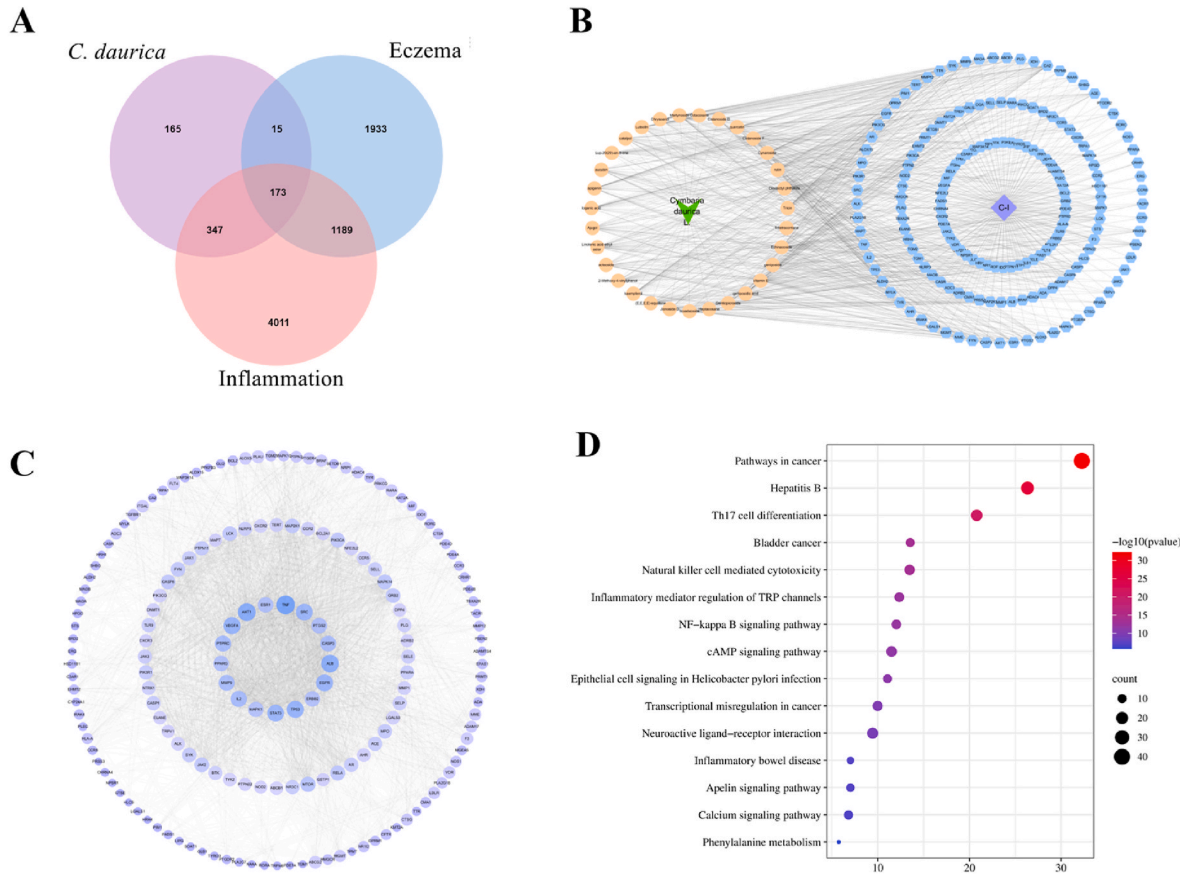

Fig. 4. Diagram of drug and disease target interactions. (A) Venn diagram of the intersection targets of *C. daurica*, eczema, and inflammation; (B) Network of compound–target interactions in *C. daurica* and eczema; (C) PPI network displaying the 173 overlapped targets of the intersection targets of *C. daurica*, eczema, and inflammation; (D) KEGG enrichment of eczema treatment with *C. daurica*.

SER120) by hydrogen bonds (Fig. 5B–D). These findings provide support for subsequent experiments.

3.3. *Animal experiments and efficacy verification*

3.3.1. *Effect of the extract of C. daurica on the DNCB-treated rat skin*

After DNCB induction, erythema, exudation, thickening, rough surface, and mossy lesions appeared on the treated skin of experimental animals in each group, indicating successful establishment of the eczema model. Exudation, redness, and swelling were reduced post treatment with different doses of *C. daurica* and positive drugs. (Fig. 6A–F). The experimental spleen index data of the rats were consistent with the skin score results. The spleen index of the model group was significantly higher than that of the control group. The spleen index was lower in the group treated with different dosages of the *C. daurica* extract (Fig. 6G) relative to the model group. These results show that the *C. daurica* extract can relieve DNCB-induced eczema in rats to some extent.

3.3.2. *The C. daurica extract protected rats from DNCB-induced eczema*

According to the H&E staining results, histopathological examination of skin lesions in the model group showed signs of hyperkeratosis of the epidermis with incomplete keratosis, thickening of the spinous layer, ulceration, and inflammatory cell scar tissue outside the spinous cell layer and inflammatory cell infiltration relative to the control group. However, the changes in the spinous layer and basal cells were not evident, and the superficial dermis was mildly inflammatory in the groups treated with low, medium, and high doses of the *C. daurica* extract, alleviating to some extent the skin inflammation caused by DNCB to some extent (Fig. 7).

3.3.3. *Effect of the C. daurica extract on inflammatory cytokine levels in rat serum*

According to the PPI network analysis, TNF and IL-2 targets may play a key role in eczema treatment with the *C. daurica* extract. Therefore, ELISA kits were used to detect TNF-α and IL-2 levels in rat serum and evaluate the effect of the *C. daurica* extract on these eczema targets. TNF-α and IL-2 levels were significantly higher in the model group than in the control group (Fig. 8). TNF-α levels in low-, medium-, and high-dose groups significantly decreased. The partial administration group showed a highly significant statistical difference ($P < 0.0001$). In addition, the level of IL-2 significantly decreased in the medium-dose group. Based on these results, we hypothesized that the extract of *C. daurica* may exert a therapeutic effect against eczema via TNF-α and IL-2 modulation (Fig. 8).

3.3.4. *Effect of the C. daurica extract on mRNA expression in rats*

RT-qPCR was used to detect the relative mRNA expression levels of key targets (SHP-1, Vav3, and Grb2) in rat skin samples to further verify the predicted related target genes in the natural killer cell-mediated cytotoxicity pathway. We found that the relative mRNA expression of the eczema model group increased when compared with the control group. However, when compared with the eczema model group, the relative mRNA expression in the low-, medium-, and high-dose treatment groups decreased. Considering the *p*-value, the differences between the groups were not significant. Based on the relative mRNA expression levels in each group, we hypothesized that the extract of *C. daurica* may play a therapeutic role in eczema by inhibiting SHP-1, Vav3, and Grb2 (Fig. 9).

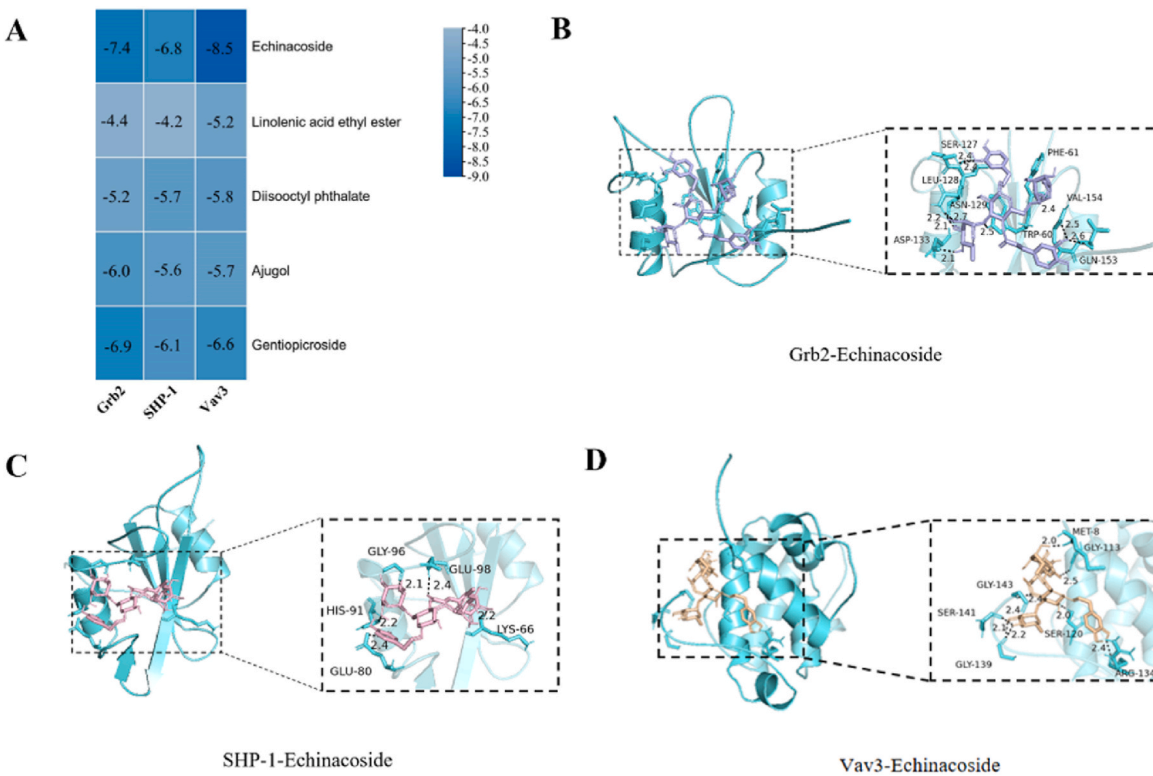

Fig. 5. Molecular docking of compounds and key targets. (A) Heat map of binding energy. The smaller the value, the stronger the combination; (B) Docking mode diagram of Grb2 with echinacoside. Grb2 is bonded with SER-127, LEU-128, ASN-129, ASP-133, PHE-61, VAL-154, TRP-60, and GLN-153 by hydrogen bonds; (C) Docking mode diagram of SHP-1 with echinacoside. SHP-1 is bonded with GLY-96, GLU-98, LYS-66, HIS-91, and GLU-80 by hydrogen bonds; (D) Docking mode diagram of Vav3 with echinacoside. Vav3 is bonded with MET-8, GLY-113, ARG134, GLY-139, SER-141, and GLY-143 by hydrogen bonds. Hydrogen bond length is indicated in the figure in Angstrom.

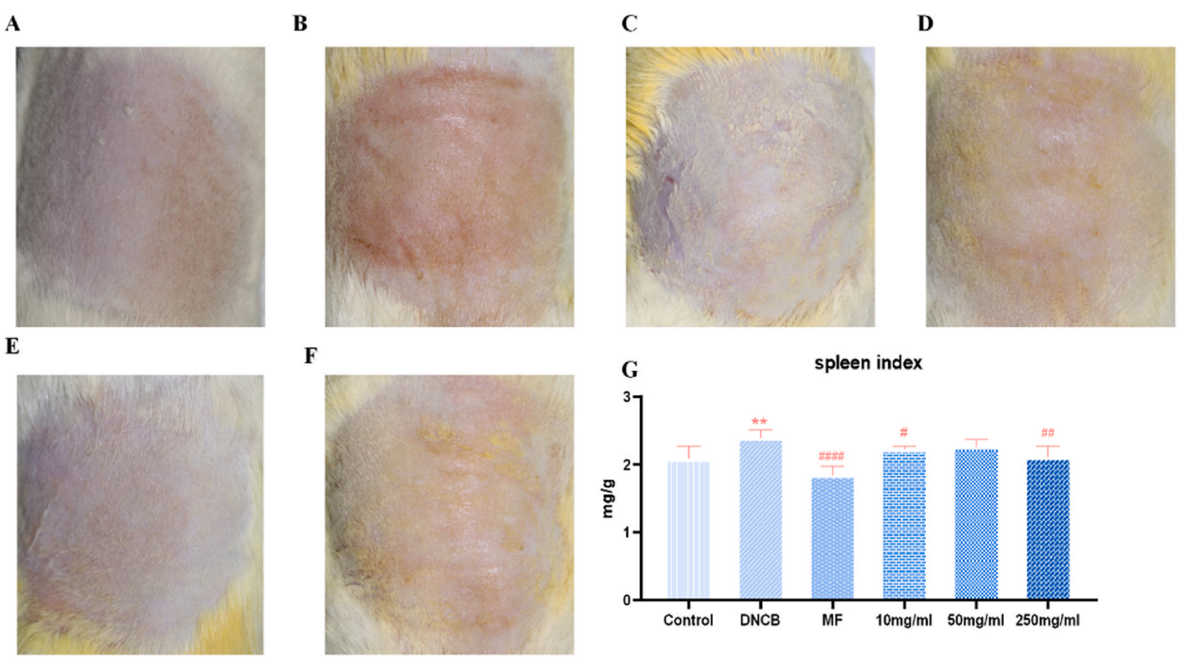

Fig. 6. Effect of *C. daurica* extract on back inflammation and spleen indices in rats: (A) normal group (Control); (B) model group (DNCB); (C) positive drug group (MF); (D) 10 mg/mL C. *daurica* group; (E) 50 mg/mL C. *daurica* group; (F) 250 mg/mL C. *daurica* group. Skin redness was used as an indicator of inflammation; whereby, increased redness indicated more severe inflammation. (G) Rat spleen index measurement results. Data are expressed as mean ± SD (n = 6). $^{**}P < 0.01$ compared with the control group; $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#\#}P < 0.0001$ compared with the model group.

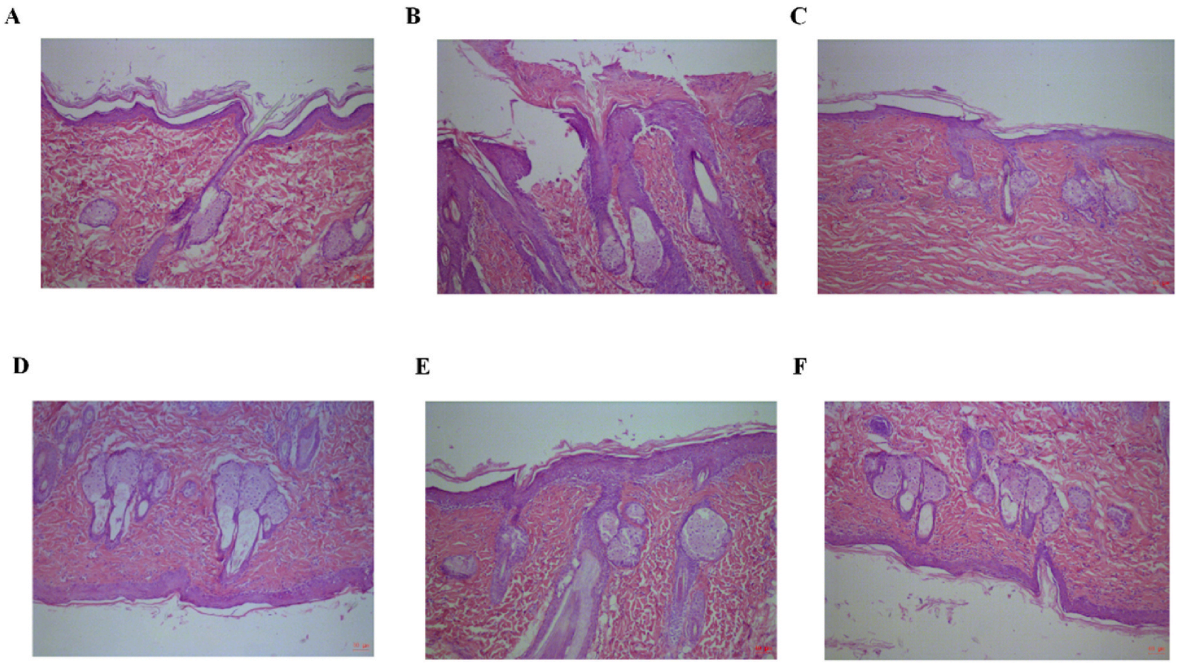

Fig. 7. H&E staining results of rats, 10 × 10 magnification, 40 μm (A) normal group; (B) model group; (C) positive drug group; (D) 10 mg/mL C. *daurica* group; (E) 50 mg/mL C. *daurica* group; (F) 250 mg/mL C. *daurica* group. Signs of epidermal hyperkeratosis with incomplete keratosis, thickening of the spinous layer, ulceration, and inflammatory cell scar tissue outside the spinous cell layer and inflammatory cell infiltration can be observed in the model group; some improvement occurred after *C. daurica* administration.

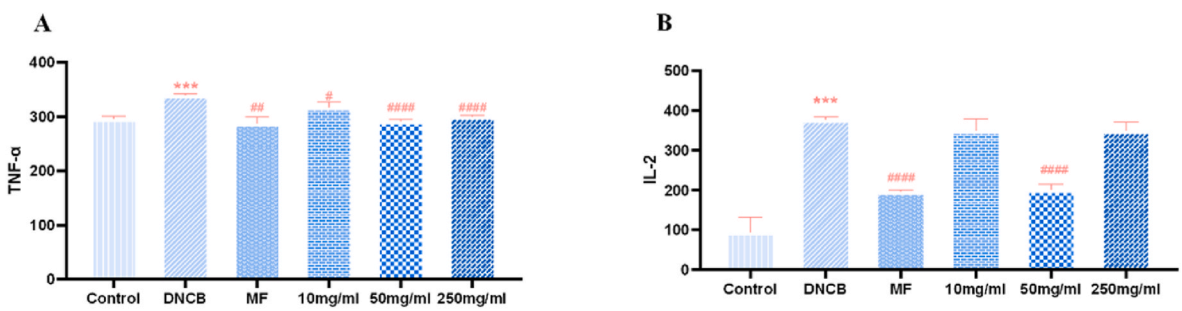

Fig. 8. Effect of *C. daurica* extract on the production of TNF-α and IL-2 in all experimental rat groups. (A) ELISA detection of TNF-α; (B) ELISA detection of IL-2. Data are expressed as the mean ± SD (n = 3). $^{***}P < 0.001$ compared with the control group; $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#\#}P < 0.0001$ compared with the model group.

3.3.5. Effect of the C. daurica extract on protein expression in rats

Western blotting was used to further verify the effect of the *C. daurica* extract on the protein expressions of SHP-1, Vav1, and Grb2 in each sample group. SHP-1, Vav1, and Grb2 expression levels were significantly higher in the eczema model group than those of the control group (Figs. 10 and 11). Compared with the model group, SHP-1, Vav1, and Grb2 protein expression was significantly lower in the medium- and high-dose treatment groups. The above results suggested that the extract of *C. daurica* may suppress the expressions of SHP-1, Vav1, and Grb2, thereby exerting anti-eczema effects, which was consistent with the mRNA expression results.

3.4. Determination of the content of ingredients in C. daurica

In view of the therapeutic effect of the *C. daurica* extract on eczema, we tried to determine content of active ingredients, which would be helpful for the subsequent quality control and application promotion of *C. daurica*. We used HPLC for the identification of verbascoside. A sample chromatogram is shown in Fig. 12. Verbascoside was separated sufficiently under the described chromatographic conditions. Calibration curves exhibited high linearity: Y = 24532X – 0.6984 (R^2 = 0.9993). This method was validated for precision (RSD = 0.34%), repeatability (RSD = 2.43%), stability (0.15%), and standard recovery rate (1.04%). These results show that the HPLC method used in this study was precise, accurate, and sensitive enough for the quantitative evaluation of verbascoside in *C. daurica*. The mass fractions of verbascoside in the 70% ethanol extract were 72.8775%, 75.9471%, 75.8030%, 71.2609%, 71.0131%, and 76.1929%, and $\bar{x} \pm s = 73.8491 \pm 2.4248$. The HPLC method developed in this study was validated for linearity, precision, repeatability, and stability.

3.5. Physicochemical properties of C. daurica extract

The 70% ethanol extract of *C. daurica* was a brown and black solid substance. The moisture content of the extract was 6.57%, 6.63% and 6.87% in the first, second and third moisture determinations, respectively. The ash content of the extract was 5.78%, 5.46% and 5.67% in the first, second and third moisture determinations, respectively.

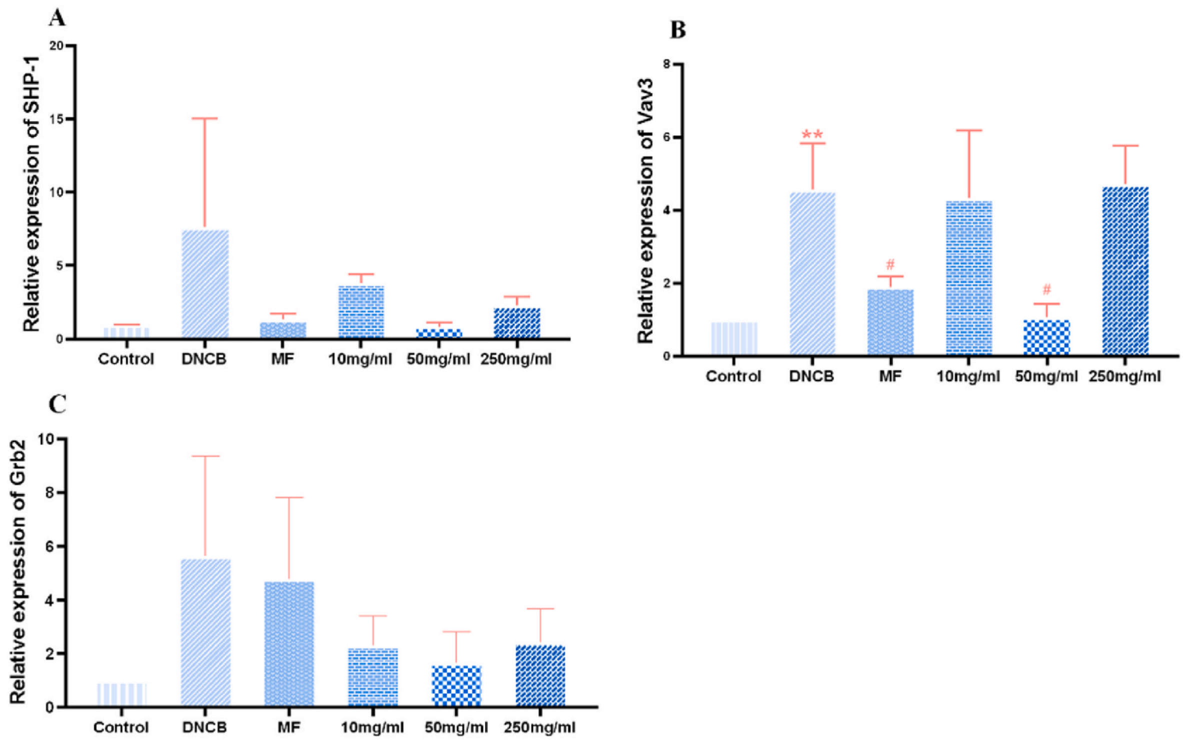

Fig. 9. Effect of the *C. daurica* extracts on the mRNA relative expression in different groups of rats. (A) mRNA relative expression of SHP-1; (B) mRNA relative expression of Vav3; (C) mRNA relative expression of Grb2. Data are expressed as the mean ± SD (n = 3). $^{**}P < 0.01$ compared with the control group; $^{\#}P < 0.05$ compared with the model group.

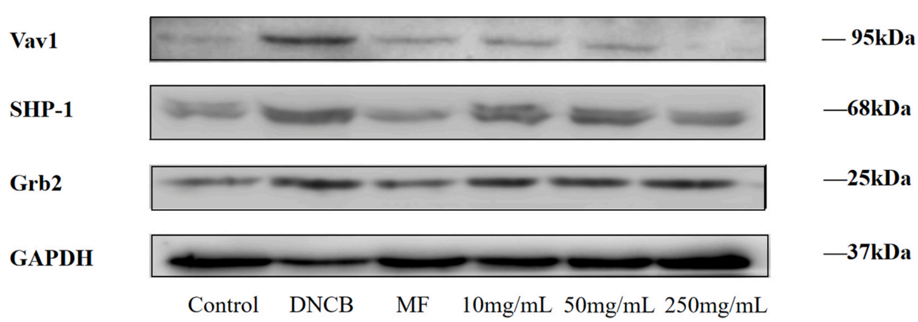

Fig. 10. Western blotting was used to detect the protein expression levels of SHP-1, Vav1, and Grb2 in all experimental rat groups.

4. Discussion

Eczema is an inflammatory skin disease (Raveendran, 2019) caused by the interaction of various genetic and environmental factors (Brown et al., 2012). The main characteristics of eczema are pruritus, recurrence, distribution, and atopy, of which the recurrence-remission process makes the disease difficult to control (Eichenfield et al., 2014). Eczema clinical diagnosis is based on symptoms such as erythema, squamous skin, pruritus, skin infiltration, and hyperkeratosis (Brown, 2016), which can occur anywhere on the body. As a skin inflammation caused by a variety of internal and external factors, eczema is prone to pleomorphic lesions and exudation (Peng et al., 2022). In addition, the skin presents impaired barrier function and inflammatory tendencies, increased skin moisture loss, dryness, and susceptibility to viruses and bacteria (Brown, 2016). Avoidance of allergens and irritants, and local or systemic anti-inflammatory therapy are commonly used in eczema treatments (Nankervis et al., 2015). Based on the high occurrence of eczema and adverse reactions experienced by patients to currently available therapeutic drugs, we explored the efficacy of the Mongolian medicinal herb *C. daurica* for treatment of eczema.

Based on its traditional use, we found that the perennial herb. *C. daurica* plays a role in the treatment of eczema; however, the mechanism is unclear. In this study, component analysis, potential mechanism analysis, the effectiveness of eczema treatment, the mechanism of eczema treatment with *C. daurica*, and the physicochemical properties of *C. daurica* were evaluated. Furthermore, the material basis and mechanism of *C. daurica* in the treatment of eczema were discussed from chemical and pharmacological perspectives in order to provide evidence for its clinical application.

The 70% ethanol extract of *C. daurica* was used in our experiment. Based on the compounds identified by LC-MS and combined with literature reports, we first conducted a network pharmacological predictive analysis, involving a total of 31 compounds and 173 targets. These compounds mainly included iridoids and alcohol glycosides. It can be considered that the role of *C. daurica* in the treatment of eczema is mainly played by these two types of substances. Among them, echinacoside and verbascoside have been studied the most, with a focus mainly on their anti-inflammatory effect. For example, Lim et al. argue that verbascoside suppressed the expression of inflammatory mediators such as inducible nitric oxide synthase, cyclooxygenase-2, nitric oxide, and

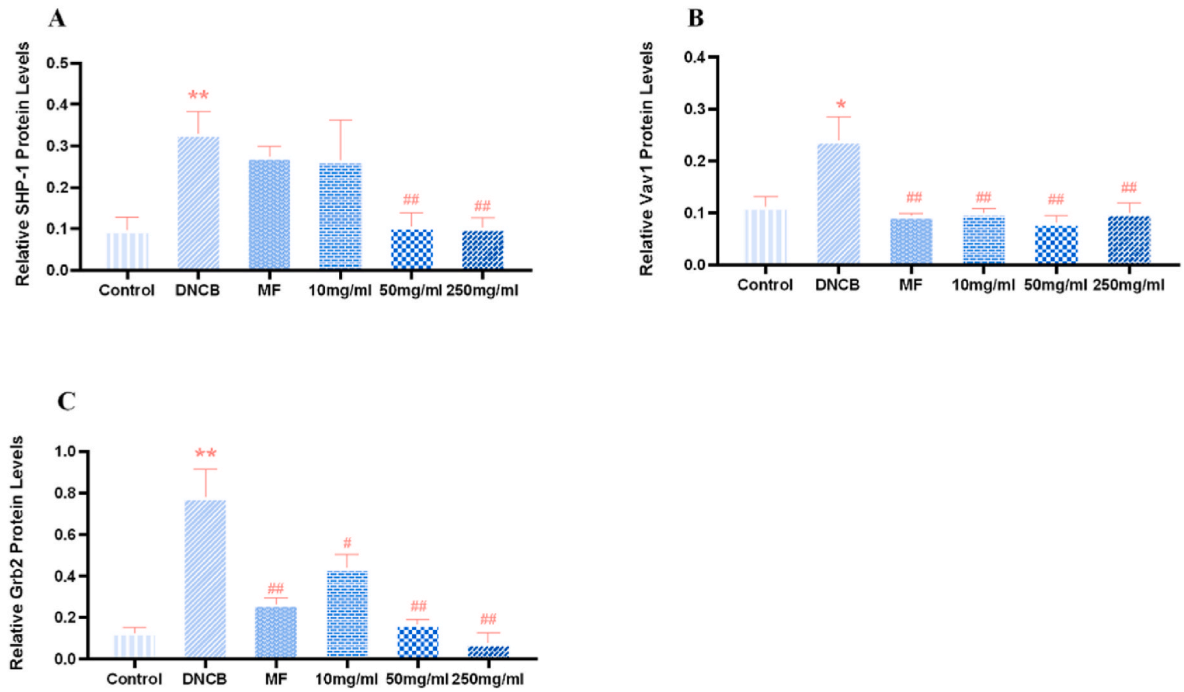

Fig. 11. Effect of *C. daurica* extract on protein expression in all experimental rat groups. (A) SHP-1 expression; (B) Vav1 expression; (C) Grb2 expression. Data are expressed as the mean ± SD (n = 3). $^{*}P < 0.05$, $^{**}P < 0.01$ compared with the control group; $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ compared with the model group.

prostaglandin E2 in the primary rat chondrocytes treated with IL-1β (Lim et al., 2021). We also determined the content of verbascoside, which is relatively high in the 70% alcohol extract of *C. daurica.* Echinacoside is another natural compound extracted from this plant that has significant biological properties, including antioxidant, anti-inflammatory, neuroprotective, anti-tumor, hepatoprotective, and immunomodulatory properties (Yang et al., 2021). In the PPI network results, the top targets were also highly associated with inflammation.

In the immune system, B cells produce antibodies, cytokines, and chemokines and carry out antigen presentation (Nagel et al., 2009). Changes in the homeostasis of B cells can lead to chronic inflammatory diseases (Daridon et al., 2009). Previous studies have shown that targeted treatment of B cell can improve atopic eczema (AE), and that TNF family B-cell-activating factor is crucial for B cell development and activation, indicating that these TNF family cytokines participate in the pathogenesis of AE (Chen et al., 2011). Meanwhile, in eczema dermatitis, the formation of spongy vesicles, an important sign of inflammatory epidermal response, was initially induced by dermis activation and the complex molecular and cellular interactions between epidermal cells and killer cells (Köck et al., 1990). Kerstan et al. demonstrated that TNF-α may prevent further inflammation at the edge of vesicle formation by lowering killer cells sensitivity threshold to CD95-induced apoptosis (Kerstan et al., 2011). TNF-α is also a key cytokine involved in the Th1 response and anti-TNF-α drugs have shown great potential in the treatment of eczema (Nakamura et al., 2017).

IL-2, a cytokine with an important role in allergic diseases, regulates T-lymphocyte mediation to maintain natural tolerance and induce tolerance, terminating lymphocyte responses by inducing inhibitory T-cell production (Christensen et al., 2006). Metabolites released after the disruption of skin epidermal homeostasis, such as IL-2, can directly activate lymphocytes (Hua et al., 2006). This is consistent with our network pharmacological analysis. In subsequent animal experiments, we also considered TNF-α and IL-2 as important indicators.

Based on KEGG analysis, the top five enriched pathways were pathways in cancer, hepatitis B, Th17 cell differentiation, bladder cancer, and natural killer cell-mediated cytotoxicity. Considering that eczema is an inflammatory skin disease and the natural killer cell-mediated cytotoxicity pathway is associated with inflammation, we selected closely related targets (SHP-1, Vav, and Grb2) in key positions of the signaling pathway map to verify the therapeutic effect of the *C. daurica* extract on eczema. SHP has been reported to inhibit toll-like receptor 4 (Kanwal et al., 2013) and NOD, LRR, and Pyrin domains containing protein 3 (NLRP3; Yuk et al., 2011), which suggests a link between SHP and inflammation (Yang et al., 2015). SHP-1 is a cytoplasmic protein and a tyrosine phosphatase family member (Dong et al., 2015) that negatively regulates innate and acquired immune responses (Neel, 1993). T cells are involved in the development of eczema, and it has been shown that T cell surface receptors (TCR) are regulated by SHP-1. Therefore, we hypothesized that the treatment of eczema could be achieved by decreasing the expression of SHP-1, improving the immune response in humans, and thus regulating the level of inflammation. The Vav protein stimulates transcription from the proximal promoter of IL-2 by activating the transcription factor and nuclear factor of activated T cells, as demonstrated in T and B lymphocytes (Cremasco et al., 2008). Therefore, we hypothesized that reducing Vav protein expression could inhibit IL-2 transcription. It has been reported that changes in T cells, mast cells, and eosinophils during allergic diseases are related to Grb2 (Lin et al., 2018). The abnormal immunoregulation in eczema includes T cell disorder, eosinophilia, and immunoglobulin secretion (Buske-Kirschbaum et al., 2005); therefore, we included Grb2 as an additional factor.

DNCB is one of the most commonly used haptens to induce chronic contact dermatitis (Kim et al., 2021). Repeated application of DNCB leads to disease characteristics similar to human atopic dermatitis (Shiohara et al., 2004). Currently, DNCB is often used to sensitize and induce eczema in animal models (Hartman et al., 1976; Gilhar et al., 2021). Therefore, we used DNCB to create a rat eczema model and conducted relevant experiments through administration and material collection in this model. The animal model was confirmed by the results of HE staining, rat spleen index, serum TNF-α, and the IL-2 index. We verified that the administration of *C. daurica* extract had some therapeutic effect on eczema in rats. Through RT-qPCR and western blotting experiments, we confirmed the results of network pharmacology and KEGG analysis; the experimental results were consistent with our

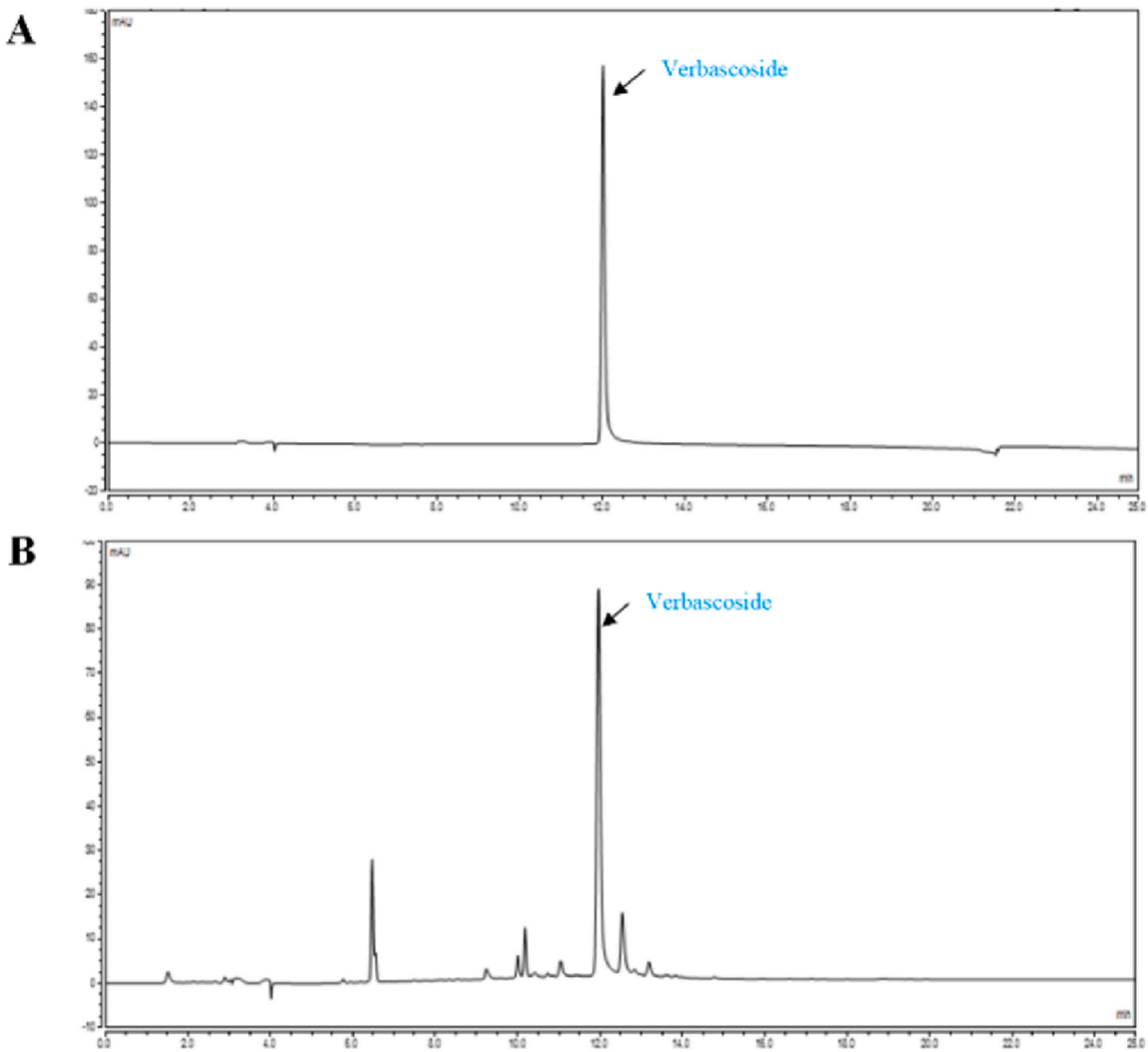

Fig. 12. HPLC pattern of (A) verbascoside and (B) verbascoside in the 70% ethanol extracts of *C. daurica*.

analysis and prediction. The main mechanism of *C. daurica* extract in eczema treatment involves reducing inflammation through the natural killer cell-mediated cytotoxicity pathway. This suggests that *C. daurica* has the potential to be developed into an effective drug to treat eczema. To better evaluate the physicochemical properties of the extract, we conducted additional water, ash, and content determination experiments with aim to provide further support for the potential medicinal applications of this plant.

5. Conclusion

This study confirmed the therapeutic effect of *C. daurica* against eczema, explored the potential mechanisms underlying the treatment of eczema, and provided a new framework for studying the pharmacological efficiency of traditional Mongolian medicine. Although only a preliminary assessment, our study highlights natural killer cell-mediated cytotoxicity as the primary mechanism of action by *C. daurica* extract in treating eczema, providing a scientific basis for future clinical research. However, we were not able to identify all active ingredients in the current study; thus, future studies should conduct comprehensive analyses of each active component and their associated mechanisms in *C. daurica* extract for treating eczema. Furthermore, the animal experiments in this study did not analyze the combination of mometasone furoate cream and 70% ethanol extract of *C. daurica*. We recommend that future studies should examine this combination to confirm potential synergistic therapeutic effects, in addition to conducting focused research on the role of *C. daurica* in eczema treatment.

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CRediT authorship contribution statement

Congying Huang: Conceptualization, Writing – original draft, Writing – review & editing. **Siqi Li:** Data processing, Interpretation of result. **Wenxin Guo:** Conceptualization, Writing – original draft. **Ziyan Zhang:** Experimental design, Animal experiments. **Xiangxi Meng:** Methodology, Experimental operations. **Xing Li:** Methodology, Experimental operations. **Bing Gao:** Data processing. **Rong Wen:** Experimental operations. **Hui Niu:** Analysis result. **Chunhong Zhang:**

Supervision, Project administration. **Minhui Li:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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